

FORM TP 2010006



TEST CODE **01212032**

JANUARY 2010

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

CHEMISTRY

Paper 03/2 – Alternative to SBA

General Proficiency

2 hours

READ THE FOLLOWING DIRECTIONS CAREFULLY.

In addition to the 2 hours allowed for the examination, candidates are allowed 10 minutes in order to read through the entire paper.

Writing may begin during the 10-minute period.

- 1. Answer ALL questions on this paper.**
- 2. Use this answer booklet when responding to the questions. For EACH question, write your answer in the space indicated and return the answer booklet at the end of the examination.**
- 3. The use of non-programmable calculators is allowed.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Answer ALL Questions.

Write your answers in the spaces provided in this booklet.

1. Students in a class conduct an experiment to determine the optimum mole ratio in which hypochlorite ions (ClO^-) react with thiosulphate ions ($\text{S}_2\text{O}_3^{2-}$) in the presence of base (OH^-).

The mole ratio of reactants that produces the maximum amount of heat will be related to the optimum mole ratio of the reactants.

The experiment involves reacting a solution of household bleach, (which contains ClO^- ions as the active ingredient) with aqueous sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$) in the presence of aqueous sodium hydroxide (NaOH), as follows.

Solution 1 is 300 cm^3 of bleach solution containing $0.50\text{ mol dm}^{-3}\text{ NaClO}$.

Solution 2 is 300 cm^3 of an aqueous solution containing 0.50 mol dm^{-3} of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ (and sodium hydroxide to make it basic).

PROCEDURE

- (a) Two measuring cylinders labelled 1 and 2 are rinsed with a small quantity of Solutions 1 and 2, respectively, and used to measure out the various volumes of Solutions 1 and 2 for each of the experiments (1 – 7) specified in Table 1.
- (b) For Experiment 1 (see Table 1 on page 4), the students followed the instructions below:
- (i) Measure 45 cm^3 of Solution 2 using Measuring Cylinder 2.
 - (ii) Pour Solution 2 from the measuring cylinder into the plastic cup provided.
 - (iii) Measure the temperature (T_1) of the solution in the plastic cup and record this in Table 1.
 - (iv) Measure 5 cm^3 of Solution 1 using the Measuring Cylinder 1.
 - (v) Pour Solution 1 from the Measuring Cylinder 1 into the plastic cup containing Solution 2.
 - (vi) Stir the solution in the plastic cup with the thermometer and record the HIGHEST temperature (T_2) reached.
 - (vii) Discard the solution in the plastic cup, rinse with distilled water and proceed with Experiments 2 – 7, using different volumes of Solutions 1 and 2 as shown in Table 1.

RESULTS

Figure 1 below gives the thermometer readings for Experiments 4 – 7. The values of T_1 and T_2 for Experiment 1 have been entered in Table 1 on page 4.

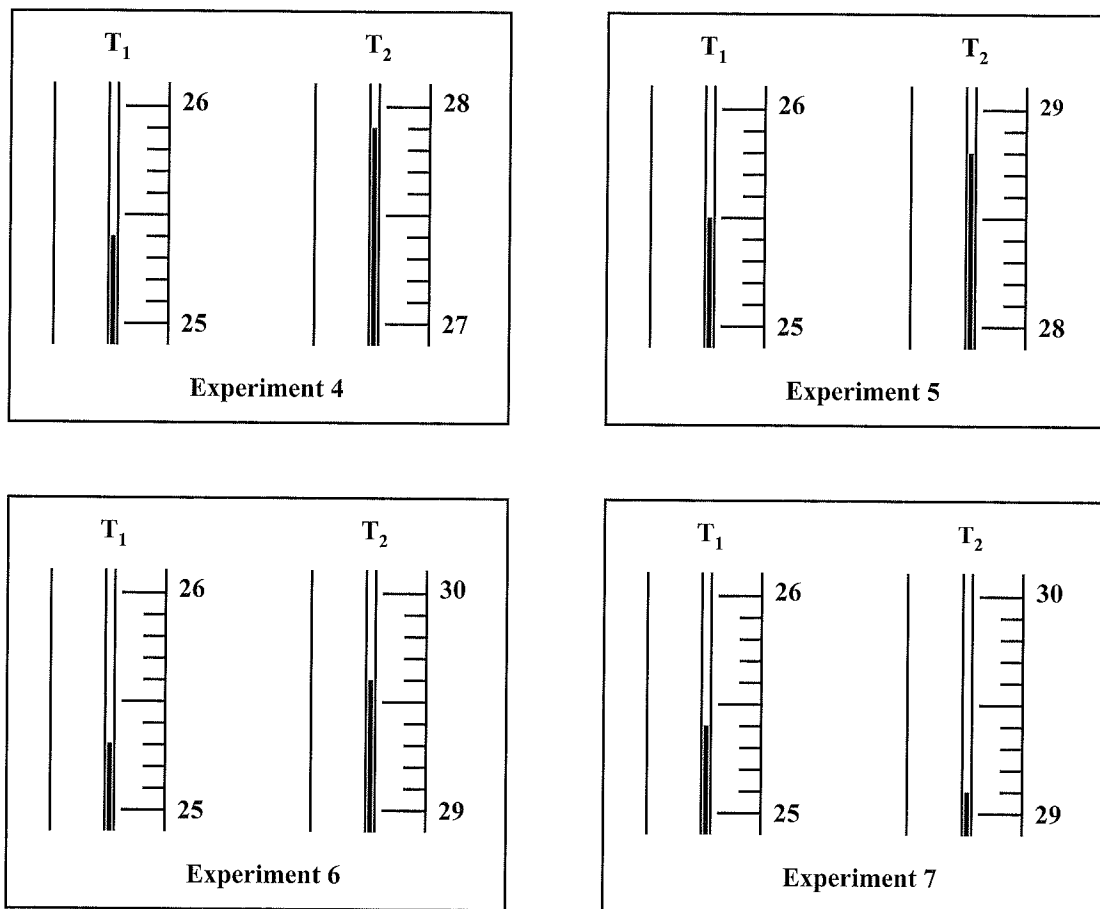


Figure 1: Thermometer readings before mixing (T_1) and after mixing (T_2) for Experiments 4 – 7

- (a) (i) For Experiments 5 to 7, take the readings of T_1 and T_2 from Figure 1, and record the values in Table 1. (6 marks)

TABLE 1: RESULTS OF EXPERIMENTS 1 – 7

Experiment	Solution 1 (cm ³)	Solution 2 (cm ³)	T_1 (°C)	T_2 (°C)	$\Delta T =$ ($T_2 - T_1$) (°C)	Volume Ratio of Solution 1: Solution 2
1	5	45	25.4	26.4	1.0	1 : 9
2	10	40	25.3	26.8		1 : 4
3	20	30	25.5	27.4		1 : 1.5
4	30	20	25.4	27.9		1.5 : 1
5	35	15				2.33 : 1
6	40	10				4 : 1
7	45	5				9 : 1

- (ii) Calculate the missing values of ΔT in Table 1. (3 marks)

- (iii) Using the axes provided on page 6, plot ΔT (°C) against experiment number. The first two points have been plotted for you. (5 marks)

- (iv) Determine the maximum value for ΔT from your graph.

(1 mark)

- (v) State the volume ratio of Solution 1 : Solution 2 when ΔT is at its maximum.

(1 mark)

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(vi) Using your answer in (v) on page 4, calculate EACH of the following when ΔT is at its maximum:

a) Number of moles of ClO^- in Solution 1

(2 marks)

b) Number of moles of $\text{S}_2\text{O}_3^{2-}$ in Solution 2

(2 marks)

(vii) Determine the mole ratio of ClO^- to $\text{S}_2\text{O}_3^{2-}$.

(1 mark)

(viii) Based on your answer in (vii) and the following unbalanced equation for the reaction, write a balanced equation for the reaction.

Unbalanced equation: $\text{ClO}^-(\text{aq}) + \text{S}_2\text{O}_3^{2-}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$

Balanced equation:

(2 marks)

(b) State TWO precautions which should be taken during the experiment.

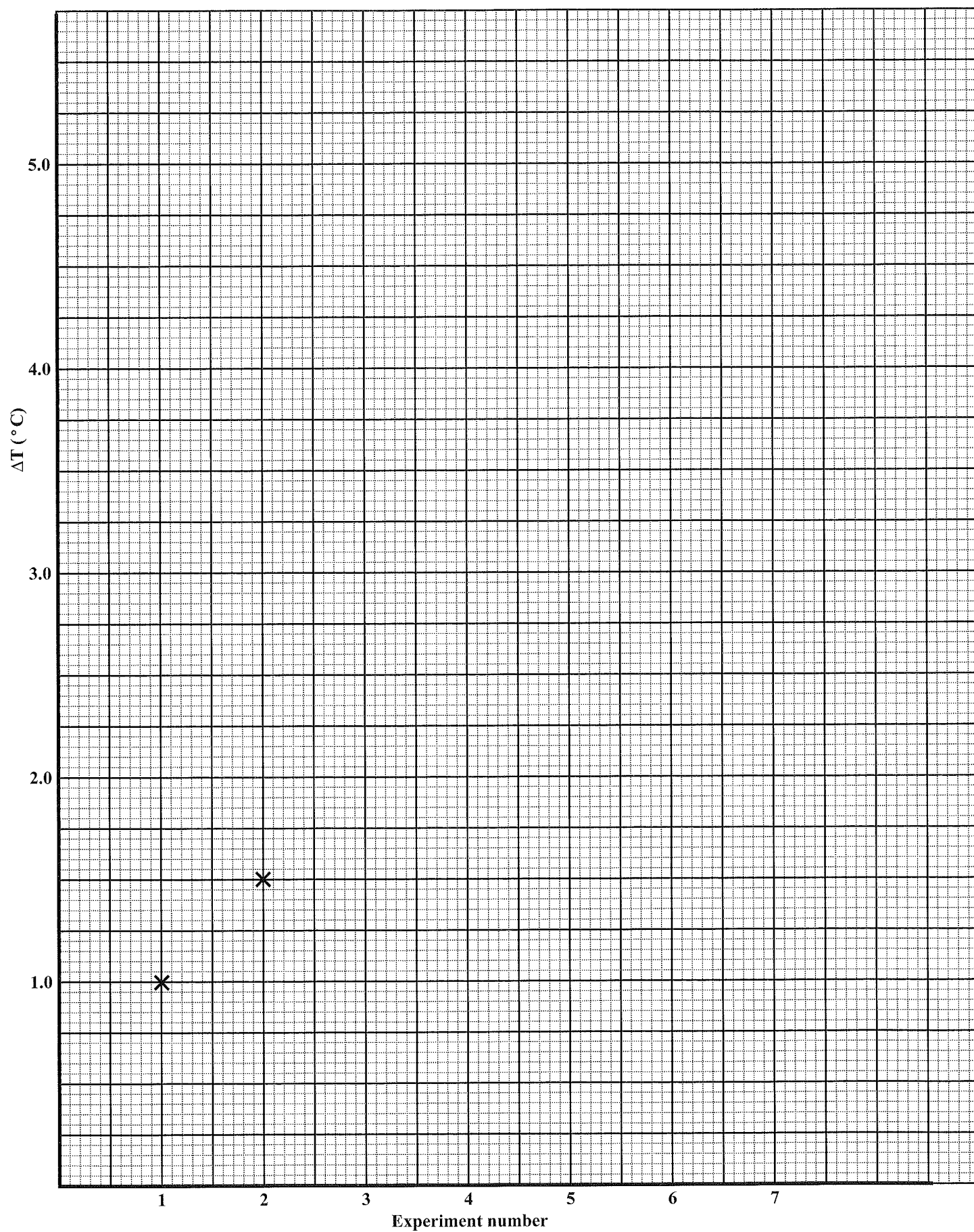
(2 marks)

(c) State ONE possible source of error.

(1 mark)

Total 26 marks

Graph for Question 1 (iii)



2. Table 2 shows the tests carried out on a Solid Y, which is a mixture of two compounds. Complete Table 2 by writing the observations that are missing.

TABLE 2: RESULTS OF VARIOUS TESTS ON SOLID Y

Test	Observation	Inference
(a) The sample of Solid Y is divided into TWO equal parts. One portion is used for Part (b) below. Dilute nitric acid is added to one portion of Solid Y. It is boiled and filtered. The filtrate is divided into THREE equal portions, for Tests (i) to (iii) below.		
(i) To the first portion of the filtrate from (a) above, aqueous sodium hydroxide is added slowly until in excess.	(2 marks)	The solution contains Al^{3+} , Pb^{2+} or Zn^{2+} ions.
(ii) To the second portion of the filtrate from (a) above, aqueous potassium iodide is added.	(1 mark)	Pb^{2+} ions are absent.
(iii) To the third portion of the filtrate from (a) above, aqueous ammonia is added until in excess.	(2 marks)	Zn^{2+} ions are present.
(b) To the second portion of the solid Y from (a) above, about 8 cm³ of water is added. It is stirred and then filtered. The filtrate is tested below.	(3 marks)	One of the compounds in the mixture is soluble in water. Sodium, potassium, or ammonium salt, or nitrate may be present.
To the portion of the filtrate from (b) above, aqueous silver nitrate is added, followed by aqueous ammonia.	(2 marks)	Chloride(Cl^-) ions are present.
		Total 10 marks

3. Plan and design an experiment to determine whether the addition of the following nitrates of Na^+ , K^+ , Ca^{2+} , Mg^{2+} and Fe^{2+} to water has the effect of making the water hard or soft.

Hypothesis: The nitrates of Na^+ , K^+ , Ca^{2+} , Mg^{2+} and Fe^{2+} will make tap water harder.

Your answer should include the following:

- (i) Apparatus and materials

(2 marks)

- (ii) Procedure

(3 marks)

- (iii) Variables to control

(2 marks)

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(iv) Data to be collected

(2 marks)

(v) Discussion of results as they relate to the hypothesis

(2 marks)

(vi) ONE possible source of error

(1 mark)

Total 12 marks

END OF TEST